

$\Delta\mathbf{W} = (0, 0)$: no standardization crossing

siren path
(GW amplitude)

source
event

$$\mathcal{H}_{\text{GW-SN}} = 2X_H = 7.44\% \\ = 0.15591 \text{ mag-equivalent}$$

observer

ladder path
(SN calibration)

$\Delta\mathbf{W} = (1, 1)$: two-endpoint gate engaged